Shadman Rabby

shadmanrabby.cse@gmail.com | Personal Website | LinkedIn | Google Scholar

Education

M.Sc. in Computer Science and Engineering, University of Dhaka *Aug 2025 – Present*

B.Sc. in Computer Science and Engineering, Khulna University of Engineering & Technology (KUET) *Jan 2019 – March 2024*

- Obtained CGPA: 3.57/4.00 (First Class)

Research Interest

Computer Vision • Trustworthy & Safe Multimodal AI • Vision-Language Model Hallucination & Robustness • Digital Media Forensics • Multimodal Learning & Reasoning

Research Experience

Copy-Move Forgery Detection Utilizing SIFT and FLANN with Optimized Keypoint Matching

- **Undergraduate Thesis** — Supervised by DR. Sk Md. Masudul Ahsan. The paper published in ICCIT 2024 [C1]. Recieved **Best Technical Presentation of the Session Award.**
- **Description:** This paper proposes a copy-move forgery detection method using SIFT for keypoint extraction and FLANN with Lowe's ratio test for reliable keypoint matching. A minimum threshold of 2 matched keypoint pairs is used to classify an image as forged, while colored patches visualize the suspected source and copied regions. Experiments on MICC-F220 and MICC-F2000 datasets achieved F1-scores of 95% and 81%, respectively. (**Paper link**)

Moral Sycophancy in Vision Language Models

- Role: First Author — Supervised by DR. Irfan Ahmad — Status: Submitted — This paper has been submitted to ACL ARR August 2026 cycle [A1]. The inference cost of the models is funded by SDAIA - KFUPM Joint research Center for Artificial Intelligence. Pre-print is available here- https://arxiv.org/abs/2602.08311
- **Description:** The study finds that VLMs often abandon correct moral judgments when users disagree, even without providing new evidence. Interestingly, models are much more likely to shift from morally right to morally wrong than the reverse, showing a clear bias toward user agreement. Open-source models are substantially more susceptible than proprietary ones.The study proposes two training-free methods, CoT+Vis and Pressure-Blind Visual Arbitration, which significantly reduce moral sycophancy. CoT+Vis achieves the strongest improvement, reducing sycophancy to below 3% on both evaluated datasets- Moralise and MM-MoralBench. .

Visual Hallucination Reduction: An Input-Level Approach for Multimodal Language Model

- Role: Second Author — Supervised by Sabbir Ahmed — Status: Submitted — This paper has been submitted to Multimedia Systems, Springer (**Pre-print version link**) [A2].
- **Description:** The study proposes a model-agnostic adaptive image preprocessing method to reduce visual hallucinations in LLMs without retraining or changing the model. By selecting noise-reduced, edge-enhanced, or original images based on the question type, it achieves a 44.3% reduction in hallucinations, improving factual grounding and reliability.

Publications

C = Conference, A = Arxiv.org

[C1.] Rabby, Shadman, and Sk Md Masudul Ahsan. "Copy-Move Forgery Detection Utilizing SIFT and FLANN with Optimized Keypoint Matching." In 2024 27th International Conference on Computer and Information Technology (ICCIT), pp. 2237-2242. IEEE, 2024.

[A1.] Rabby, Shadman, Md Hefzul Hossain Papon, Sabbir Ahmed, Nokimul Hasan Arif, A. B. M. Rahman, and Irfan Ahmad. "Moral Sycophancy in Vision Language Models." arXiv preprint arXiv:2602.08311 (2026).

[A2.] Arif, Nokimul Hasan, Shadman Rabby, Md Hefzul Hossain Papon, and Sabbir Ahmed. "Visual Hallucination Reduction: An Input-Level Approach for Multimodal Language Model." (2025).

Papers in Preparation

- Md.Al-Amin-Al-Noor, Shadman Rabby, *Evaluating the Robustness of Large Vision-Language Model–Powered Autonomous Agents Against Adversarial Spam Attacks in Computer Task Execution.* [In progress]
- Shadman Rabby, Chowdhury Farhan Ahmed *Cross-Modal Evidence Reasoning for Multimodal Fake News Detection in Video News.* [In progress]
- Siddik Howlader, Shadman Rabby, *Cross-Linguistic Validation of Dyslexia Simulation via Visual-Word-Form Units: A Bangla Extension* [In progress]

Professional Experience

Lecturer, Department of CSE, Daffodil International University(DIU) *July 2024 – Present*

- Supervising five students (one student has completed graduation)
- Mentoring 67 students
- Course Taught: Algorithm, Compiler Design, Programming and Problem Solving, Database Management System Digital logic Design, Microprocessor and Microcontoller (Both theory and Lab)

Technical Skills

Programming Languages	C, C++, Python, Java
Data Analysis	Scikit-learn, PyTorch
Web Programming	HTML, CSS, PHP, JavaScript
Databases	MySQL
Miscellaneous	Git, LaTeX, Kaggle, NotebookLM

Achievement & Scholarship

- **2024 ICCIT Best Technical Presentation of the Session Award**
  2024 27th International Conference on Computer and Information Technology (ICCIT)
  20-22 December 2024, Cox's Bazar, Bangladesh 2024
- **Dean's List Award (Offered to average GPA of 3.75 or above achievers)**
  Received Dean's Award for achieving average GPA 3.75 out of 4.00 2019-20
- **Country Rank: 3^{rd}, Global Rank: 115^{th} (out of 2403 teams)**
  IEEEXtreme 15.0 Programming Competition arranged by IEEE 2022

Major Projects

OMR Sheet Evaluator Website
OMR system that processes uploaded sheets using filtering, thresholding, and morphological operations to extract MCQ answers, supporting single/multiple uploads and storing results by student ID using Python, OpenCV, and a web-based stack (JS, PHP, MySQL).

Finding Best Hospital
Android application to compare the services of the registered hospitals/clinics using Java and Firebase

Fruit Slicing Game
A fruit-slicing 3D game developed using OpenGL v3.3

Co-Curricular Activities

- **Advisor**
  Computer Programming Club (CPC), ACM wing, Daffodil International University 2024 - Present
- **Member**
  Member of Publication and Co-ordination wing, NLP & ML Lab, DIU. 2024 - Present
- **Performance analyzer**
  Special Group Interested in Programming Contest (SGIPC), KUET 2020 - 2023
- **Ambassador, IEEE Student Branch KUET,** IEEEXTREME 15.0, IEEE 2021
- **Problem Solving from different online judges:**

Codeforces	shadman85 -570+ solved
AtCoder	shadman85 -63+ solved
Vjudge	shadman99 -216+ solved